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MODULAR ARTIFICIAL POOL, SUCH AS A MODULAR SWIMMING POOL, AND METHOD OF PRODUCING SUCH A POOL

Abstract

The invention concerns an artificial pool, the artificial pool comprising: a plurality of panels arranged along the perimeter of the artificial pool, two successive panels being removably fixed to one another to form the lateral walls of the artificial pool; a plurality of lateral uprights removably fixed to at least one panel; for each of the lateral walls of the artificial pool at least one respective first crossmember arranged along said lateral wall and fixed to each panel of said lateral wall. The artificial pool further comprises first angle irons to which the first crossmembers are removably fixed to form a first frame.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of European Application No. EP24305318.8, filed Feb. 29, 2024, the entire disclosures of which are hereby incorporated herein by reference.

TECHNICAL FIELD

[0002] The invention concerns the field of artificial pools, such as that of swimming pools.

[0003] The invention has more particularly for object an artificial pool notably of the swimming pool type and a method of producing such a pool.

PRIOR ART

[0004] Equipping a building with an artificial pool, such as a swimming pool, is relatively long and costly. In fact, in addition to the groundworks for forming the excavation in which the artificial pool will be arranged, it is necessary to employ a mason to pour a slab that will form the bottom of the artificial pool and to erect the lateral walls of the pool. It is only after the slab and these lateral walls have set that it is possible to equip the pool by installing a coating (generally a liner or tiles) and providing a system for filtering the water. Because of the relatively long drying times, the installation of the swimming pool can take up to three months.

[0005] In order to remedy these drawbacks, without having to employ the lower quality and less adaptable solutions of composite shell swimming pools, it is known to produce modular pools from metal panels. Such pools are generally made based on a welded metal framework onto which are welded panels forming the lateral walls, the panels themselves being welded to one another. These welds can be produced in the factory, necessitating exceptional transportation of the artificial pool or at least part of it to the installation site, necessitating equipment and specialist technicians to travel to the site.

[0006] Nevertheless, although such a solution enables a considerable saving of time compared to the concrete artificial pools referred to above and possibly making it possible for some of the production time to take place in the factory when the welds are made there, the production times remain relatively long and can necessitate the use of exceptional transportation for moving the artificial pool from the factory to the installation site.

STATEMENT OF INVENTION

[0007] The invention aims to remedy at least some of the above drawbacks and thus has in particular for object providing an artificial pool which, while having the modular design and the quality of concrete artificial pools and modular artificial pools with metal panels, can easily and rapidly be installed on site without necessitating exceptional transportation.

[0008] To this end the invention concerns an artificial pool, notably of the swimming pool type, the artificial pool having at least four lateral walls, two successive lateral walls being connected to one another by an edge, the artificial pool comprising: [0009] a plurality of panels arranged along the perimeter of the artificial pool, two successive panels being removably fixed to one another in such a manner that the panels of said plurality of panels together form the lateral walls of the artificial pool, [0010] a plurality of lateral uprights arranged along the perimeter of the artificial pool, each lateral upright being removably fixed to at least one panel and preferably being fixed to the at least one panel at the level of a junction with another panel, [0011] for each of the lateral walls of the artificial pool, at least one respective first crossmember arranged along said lateral wall, said first crossmember being fixed to each panel forming part of said lateral wall either via the lateral upright fixed to said panel or via a fixing part removably fixed to said panel, [0012] the artificial pool further comprising first angle irons, the first crossmembers being removably fixed together at the level of the edges by said first angle irons in such a manner as to form a first frame.

[0013] The combination of a removable assembly, such as for example a screwed or bolted assembly, and modular elements in the form of the panels and the lateral uprights, allow the production of an artificial pool the dimensions of which can easily be adapted while allowing rapid installation on site. In fact, such a pool does not necessitate a long step of drying the lateral walls, as is the case with concrete pools, or complex assemblies such as by welding a framework, welding plates to one another or welding plates to such a frame. Removable assembly is generally much simpler and can be carried out simultaneously by a number of not necessarily specialist workers. The invention therefore opens up the possibility of installing a modular pool in a few days.

[0014] Furthermore, this use of a removable assembly is not to the detriment of the solidity and the quality of the artificial pool, since the first frame, formed by the first crossmembers, is fixed by the lateral uprights, and possibly fixing parts, to each panel, and therefore allows good recovery of forces. Because of this, the risks linked to the pressure of the water intended to be contained in said artificial pool are contained, even eliminated, and the artificial pool therefore has a solidity similar to that of pools based on welded panels.

[0015] Thus the invention enables the production of an artificial pool benefitting from the adaptability advantages offered by concrete artificial pools and modular artificial pools using welded metal panels while allowing particularly rapid installation without necessitating use of exceptional transportation from the factory to the installation site.

[0016] Finally, the use of such a removable fixing renders the artificial pool in accordance with the invention easily recyclable. In fact, in contrast to prior art artificial pools, either made of concrete, and therefore difficult to recycle, or consisting of welded panels, and thus necessitating cutting operations, the artificial pool in accordance with the invention necessitates only removal of the removable fixing elements to separate the assemblies of elements constituting the pool. These elements are then perfectly able to be used again to form a new artificial pool, only the coating having to be changed, or easily transferred to appropriate recycling facilities.

[0017] By removably fixed is meant hereinabove and in the remainder of the present document fixed by a reversible mechanical assembly such as screws, rivets, bolts. Such removable fixing excludes definitive or semi-definitive fixing by welding and gluing.

[0018] The artificial pool may further comprise for each lateral wall at least one respective second crossmember arranged along said lateral wall parallel to the first crossmember corresponding to said lateral wall, said second crossmember being fixed to each panel forming part of said lateral wall either via the lateral upright fixed to said panel or via a fixing part removably fixed to said panel, [0019] the second crossmembers being connected to one another at the level of the edges either by the first angle irons or by second angle irons of the artificial pool in such a manner as to form a second frame parallel to the first frame.

[0020] With such second crossmembers and the second frame that they allow to be formed, the artificial pool features a recovery of forces applied to the panels and therefore good resistance to the pressure of the water intended to be contained in the pool. Because of this, the risks linked to the pressure of the water intended to be contained in said artificial pool are perfectly contained, even eliminated, and because of this the artificial pool has the optimum solidity.

[0021] Each lateral upright may be intended to be fixed to the ground.

[0022] Such fixing to the ground enables transfer of some of the forces exerted on the panels to ground level, favoring the solidity of the present pool.

[0023] The artificial pool may further comprise at least one support strut element removably fixed to at least one panel, the artificial pool preferably comprising four lateral walls, two longitudinal walls and two walls transverse to said longitudinal walls, and the artificial pool comprising for each longitudinal wall at least two support strut elements.

[0024] Such support strut elements enable transfer to the ground of some of the horizontal forces on the panels, such as those exerted by the water intended to be contained in the pool. In this way the pool has good stability relative to the pressure exerted by said water.

[0025] The artificial pool may comprise for at least one of the lateral walls at least two support

struts arranged along said lateral wall, and the artificial pool further comprising for said lateral wall at least one secondary crossmember, said secondary crossmember being fixed to said at least two support struts.
[0026] Such secondary crossmembers enable good distribution of forces on the support strut elements. These secondary crossmembers can therefore contribute to the good pressure resistance of the pool in accordance with the invention.
[0027] The artificial pool may further comprise a platform accommodated at least in part in a volume delimited by the at least four lateral walls, said platform being mounted to be mobile between a first, so-called low position and a second, so-called high position in such a manner as to modify an apparent depth of said artificial pool.
[0028] Such a platform enables production of a variable depth pool, thus offering the possibility of adapting the depth of the pool depending on the intended activity in the latter and the persons wanting to use it. In accordance with one possibility of the invention, such a platform can also be used, in the high position, as an element for closing the pool. Thus, given this possibility, the platform in this high position can participate in the formation of a patio both enabling making the pool safe and allowing the use of the space occupied by the pool for other purposes when the pool is not occupied.
[0029] The artificial pool may further comprise: [0030] a system for actuating the platform arranged outside the volume delimited by the at least four lateral walls, [0031] at least four cables connecting the platform to the actuating system, movement of said cables being driven by the actuating system, and at least four pulleys arranged along the lateral walls in such a manner as to guide a respective cable to apply vertical traction to the platform, the actuating system preferably comprising a cylinder the two ends of travel of which respectively correspond to the first position and the second position.
[0032] The artificial pool may comprise an upper frame fixed to the panels on an upper part of the latter, the upper frame having two longitudinal sides and two lateral sides, each longitudinal and/or lateral side of the upper frame, preferably each longitudinal side, has a section of U general shape accommodating between the arms of the U rolling members such as pulleys, and said frame being configured to allow: [0033] removable fixing of the upper frame to the panels by the bottom of the U, the rolling elements then being adapted to provide a support for a sliding cover, and [0034] removable fixing of the upper frame to the panels by its arms, the artificial pool being a pool according to the invention and comprising the platform, the rolling elements forming the at least four pulleys arranged along the lateral walls in such a manner as to guide the respective cable.
[0035] Such an upper frame offers the possibility of using the same upper frame either for winding or unwinding a sliding cover of the pool or to form a mobile bottom conforming to the present embodiment. Thus it is perfectly possible to adapt the configuration of the artificial pool without having to provide a plurality of separate upper frames.
[0036] Such an actuation system enables movement of the platform in a manner perfectly integrated into the artificial pool in accordance with the invention.
[0037] The panels can be identical and each panel advantageously has: [0038] a width between 25 and 100 cm, even between 40 and 75 cm, or again between 45 and 55 cm, this width being particularly advantageously substantially equal to 50 cm, [0039] a height between 50 and 220 cm, even between 75 and 190 cm, the height advantageously being selected from 80, 100, 120, 140, 160 and 180 cm.
[0040] The use of identical panels renders the pool in accordance with the invention entirely modular, only the number of panels and the length of the crossmembers adapted to obtain the pool with the required dimensions having to change.
[0041] Each panel may comprise a bent plate at least two longitudinal edges of which are bent, two successive panels of a lateral wall being assembled to one another by means of their longitudinal edges.

[0042] Such bends favor the assembly of the panels with the lateral uprights and the fixing parts, if any.
[0043] The invention also concerns a method of assembling an artificial pool, notably of the swimming pool type, the artificial pool having at least four lateral walls, two successive lateral walls being connected to one another by an edge, said method comprising the following steps:
[0044] providing a plurality of panels intended to be arranged along the perimeter of the artificial pool and together to form the lateral walls of the artificial pool, [0045] providing a plurality of lateral uprights intended to be arranged along the perimeter of the artificial pool, [0046] providing for each lateral wall of the artificial pool at least one respective first crossmember intended to be arranged along said lateral wall, [0047] providing first angle irons, [0048] arranging the panels along the perimeter of the artificial pool and removably fixing the panels to one another in such a manner that two successive panels are removably fixed to one another and the panels of said plurality of panels together form the lateral walls of the artificial pool, [0049] arranging the lateral uprights along the exterior perimeter of the artificial pool and removably fixing each lateral upright to at least one panel, said lateral upright preferably being fixed to the at least one panel at the level of a junction with another panel, [0050] arranging each first crossmember along said corresponding lateral wall and fixing said first crossmember to each panel constituting said lateral wall either via the lateral upright fixed to said panel or via a fixing part removably fixed to said panel, the first crossmembers being removably fixed to one another at the level of the edges by the first angle irons in such a manner as to form a first frame.
[0051] Such a method enables an artificial pool in accordance with the invention to be produced and the benefit of the advantages associated therewith to be obtained.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0052] The present invention will be better understood after reading the description of embodiments provided by way of non-limiting illustration only with reference to the appended drawings, in which:
[0053] FIG. **1** depicts an artificial pool in accordance with a first embodiment of the invention,
[0054] FIG. **2** depicts an artificial pool in accordance with a second embodiment having a variable depth,
[0055] FIGS. **3**A and **3**B are respectively a view from above and a side view of the artificial pool in a variant of the artificial pool depicted in FIG. **2**,
[0056] FIG. **4** depicts an actuator of the artificial pool depicted in FIGS. **3**A and **3**B,
[0057] FIGS. **5**A to **5**D are respectively a perspective view of an upper frame of the artificial pool depicted in FIGS. **3**A and **3**B, a close-up perspective view of a pulley element of such a frame, a section of the same pulley element, and a view in section of this frame and a mobile bottom, said upper frame integrating a transmission system between the actuator and a mobile bottom of the pool,
[0058] FIGS. **6**A to **6**C are respectively a perspective view of the mobile bottom of the artificial pool depicted in FIGS. **3**A and **3**B, a close-up view of a corner of said mobile bottom and a close-up view in section at the level of a traction point of said mobile bottom.
[0059] Identical, similar or equivalent parts in the various figures bear the same reference numbers so as to facilitate passing from one figure to another.
[0060] The figures are clarified because the various parts represented in the figures are not necessarily represented to the same scale.
[0061] The various possibilities (variants and embodiments) must be understood as not mutually exclusive and may be combined with one another.

DETAILED DISCLOSURE OF PARTICULAR EMBODIMENTS

[0062] FIG. **1** depicts a perspective view of a modular artificial pool **100** in accordance with the invention.

[0063] While such an artificial pool **100** in accordance with the invention can in particular be an artificial pool of the swimming pool type, it can obviously be of some other type such as a pond in a garden.

[0064] In accordance with the invention, the artificial pool **100** has at least four lateral walls **101**, **102**, two successive lateral walls **101**, **102** being connected to one another by an edge **103**. In a usual configuration of the invention the artificial pool **100** includes four lateral walls **101**, **102**, including two longitudinal walls **101** and two walls **102** transverse to said longitudinal walls **101**. Of course, the invention is not limited to only this configuration and the artificial pool **100** in accordance with the invention may comprise six to eight lateral walls **101**, **102** without departing from the scope of the invention.

[0065] It will be noted that in a usual configuration of the invention the bottom of the artificial pool **100** may be formed by a slab **106** shown in FIG. **2**. Alternatively, the bottom of the artificial pool **100** may be formed by a floor, for example by one or more metal plates or by some other support that is not shown. In accordance with this possibility, as is also shown in FIG. **2**, the bottom may comprise a set of beams **107** serving as a base for the metal plates or other supports. In this variant in which the bottom of the artificial pool **100** is not formed by a concrete slab the artificial pool **100** may be formed on some other type of surface, such as a bed of sand or a beaten and stabilized earth surface.

[0066] Likewise, it will be noted that when a horizontal or vertical direction is referred to, or more generally a horizontal or vertical qualifier is used, this direction or this qualifier is relative to the elements of the artificial pool as they are oriented when the artificial pool **100** is installed on site, it being understood that the lateral walls **101**, **102** of the artificial pool **100** are vertical and that the bottom is horizontal. Thus if, for example, it is mentioned hereinafter that the lateral uprights **120** extend vertically, it is to be understood that these lateral uprights **120** are vertical when they equip the artificial pool **100**, being installed along the vertical walls and extending parallel to the latter and perpendicularly to the bottom. Thus, if these elements are not equipping the artificial pool **100**, for example before assembling them to form the artificial pool **100**, the qualifiers horizontal and vertical do not apply and these elements may have any other orientation.

[0067] As shown in FIG. **1**, the artificial pool **100** comprises: [0068] a plurality of panels **110** arranged along the perimeter of the artificial pool **100**, two successive panels **110** being removably fixed to one another in such a manner that the panels **110** of said plurality of panels together form the lateral walls **101**, **102** of the artificial pool **100**, [0069] a plurality of lateral uprights **120** arranged along the perimeter of the artificial pool **100**, each lateral upright **120**, **121** being removably fixed to at least one panel **110** and fixed to the at least one panel **110** at the level of a junction with another panel **110**, [0070] for each of the lateral walls **101**, **102** of the artificial pool **100** a first respective crossmember **131** arranged along said lateral wall **101**, **102**, said first crossmembers **101**, **102** being fixed to each panel **110** forming part of said lateral wall **101**, **102** either via the lateral upright **120**, **121** fixed to said panel **110** or via a fixing part **123** removably fixed to said panel **110**, [0071] for each lateral wall **101**, **102** a second respective crossmember **133** arranged along said lateral wall **101**, **102** parallel to the first crossmember **131** corresponding to said lateral wall **101**, **102**, said second crossmember **133** being fixed to each panel **110** forming part of said lateral wall **101**, **102** either via the lateral upright **120**, **121** fixed to said panel **110** or via a fixing part **122** removably fixed to said panel **110**.

[0072] The artificial pool **100** further comprises first angle irons **140**, the first crossmembers **131** being removably fixed together at the level of the edges **103** by said first angle irons **140** in such a manner as to form a first frame **132**. In the same manner, the second crossmembers **133** are removably fixed together at the level of the edges **103** by these same first angle irons **140** in such a

manner as to form a second frame **134** parallel to the first frame **132**.
[0073] Of course, if in a usual configuration the pool comprises two frames **132**, **134**, one of which is a lower frame near a lower part of the lateral walls **101**, **102** and the other is an upper frame near an upper part of the lateral walls **101**, **102**, it may equally well comprise a single frame between the lower part and the upper part of the lateral walls **101**, **102** or three or even more frames, without this departing from the scope of the invention. Note likewise that, although the first crossmembers **131** and the second crossmembers **133** are fixed by the same first angle irons **140**, without departing from the scope of the invention and in accordance with a possibility that is not depicted, it is perfectly feasible to envisage the second crossmembers **133** being connected together by second angle irons distinct from the first angle irons **140**.
[0074] As shown in FIG. **1**, each panel **110** comprises a rectangular plate. To be more specific, each panel **110** may be a bent plate two longitudinal edges of which are bent in such a manner as to allow assembly of two successive panels **110** of a lateral wall **101**, **102** by means of their longitudinal edges. Each panel **110** also has two bent lateral edges transverse to the longitudinal edges. All the bends of the longitudinal and lateral edges are made in the same direction in such a manner that the panels **110** have their bends oriented toward the exterior of the artificial pool **100** during assembly thereof to form the lateral walls **101**, **102**.
[0075] In order to allow removable assembly of the panels, for example by means of bolts, the longitudinal edges of the panels may be perforated. In exactly the same manner and in particular, as is the case for the artificial pool **100** in accordance with the second embodiment described later with reference to FIG. **2**, to enable assembly of an upper frame the lateral edges may be perforated. In order to reinforce the panels **110**, as shown in FIG. **1**, the latter may be equipped with brackets **110**A connecting together the longitudinal and lateral edges.
[0076] Accordingly, in the present embodiment, as shown in FIG. **1**, for each lateral wall **101**, **102** each panel **110** is assembled to the next panel along said lateral wall **101**, **102** by one of its longitudinal edges, said longitudinal edge being assembled to the longitudinal edge of the next panel by a removable fixing system, such as by nuts and bolts arranged in the perforations formed along the longitudinal edges.
[0077] In terms of dimensions, the panels **110** each have the following dimensions: [0078] a width between 25 and 100 cm, even between 40 and 75 cm, or again between 45 and 55 cm, this width being particularly advantageously substantially equal to 50 cm, [0079] a height between 50 and 220 cm, even between 75 and 190 cm, the height advantageously being selected from 80, 100, 120, 140, 160 and 180 cm.
[0080] Accordingly, in the present embodiment the panels have a width of 50 cm and a height of 180 cm. In this way the artificial pool **100** depicted in the figure has a width of 3 m, a length of 8 m and a depth of 1.8 m.
[0081] In addition to this assembly with the next panel **110**, each panel **110** is itself assembled to at least one lateral upright **120** and at least one fixing part **123**, namely, in the present embodiment, two fixing parts **123** one associated with the first crossmember **131** and the other associated with the second crossmember **133**. Each panel **110** is preferably removably fixed to a lateral upright **120** or a fixing part **123**, **140** at the level of a junction between two successive panels **110**, that is to say the longitudinal edges of these two successive panels **110**. To be more specific, the fixing of the lateral upright **120** or the fixing parts **123** to these two successive panels **110** can advantageously be carried out during the assembly of these two panels **110**, the lateral upright **120** or the fixing parts **123** then being assembled to these two panels **110** on an internal side of a longitudinal edge of one of these two panels **110**.
[0082] In the present embodiment, the lateral uprights **120** take the form of a U-section with one of the arms of the U extended by flanges defining between them passages for the first and second crossmembers **131**, **132** via which they are fixed to the panel **110**. In this way when each vertical lateral upright **120** is fixed to a panel **110** by its flanges the gap between the two flanges forms a

space to receive either the first crossmember **131** or the second crossmember **133**. As discussed hereinafter with reference to the second embodiment and FIG. **2**, the U-section is able to accommodate guide elements of an upper frame **153**.

[0083] Each fixing part **123** has a U-shape or is stirrup shaped, with a bottom and two arms, and is shaped to be removably fixed to a panel **110**. In this way, after fixing the fixing part **123** to the corresponding panel **110** the space delimited between the bottom and the two arms of the fixing part **123** and the longitudinal edge of the panel **110** forms a passage for either the first or the second crossmember **131**, **133**.

[0084] In addition to the lateral uprights **120** and the fixing parts **123**, each of the first and second crossmembers **131**, **132** is also removably fixed to the two angle irons **140** of the corresponding lateral wall **101**, **102**. Each angle iron **140** has a longitudinal general shape, with a V-shape lateral section, each of the arms of the V including rectangular openings at least a part of which forms a gap to receive either the first crossmember **131** or the second crossmember **132**. The lateral and exterior longitudinal edges of the arms of the V are bent in a direction exterior to the V. In a similar manner to the lateral uprights **120**, each angle iron **140** has perforations to enable fixing of said angle iron **140** to the first and second crossmembers **131**, **133** and to the panels **110** corresponding to said edge **103**.

[0085] In order for the artificial pool **100** to have a high resistance to forces exerted by the water that it is intended to contain, the artificial pool **100** may comprise support strut elements **121**. To be more specific, and in a usual configuration of the invention, the artificial pool **100** may comprise for each longitudinal wall **101** at least two support strut elements **121**, in the present embodiment and as shown in FIG. **1** three support strut elements **121**.

[0086] Each support strut element **121** takes the form of a right-angle triangular metal plate at least two main edges of which are bent, namely the edge corresponding to the hypotenuse and the horizontal edge. To be more specific, to provide good force transfer in a direction oblique to the vertical the side corresponding to the hypotenuse has a U-shape double bend. The horizontal edge includes perforations in order to enable fixing of the support strut to the ground.

[0087] As depicted in FIG. **1**, the support strut elements **121** may be of two types, a first type of support strut element **121**A shaped to be fixed to a vertical upright **120** and a second type of support strut element **121**B shaped to be associated with the fixing parts **123**.

[0088] The first type of support strut element **121**A has its vertical edge bent and includes perforations along said bend to enable it to be fixed to the lateral upright **120**.

[0089] Like the vertical uprights **120**, the second type of support strut element **121**B has its vertical edge extended by flanges defining between them passages for the first and second crossmembers **131**, **132**. These flanges include perforations in order to enable them to be fixed to the longitudinal edges of the panels **110**.

[0090] Whether the support strut element **121** is of the first type or the second type, it has along its surface in a vertical direction passages for one or more secondary crossmembers **135**, in the present embodiment a single secondary crossmember **135**, so as to enable transfer of forces between the support strut elements **121** of the same longitudinal wall **102**. Thus in the present embodiment the artificial pool comprises two secondary crossmembers **135** each associated with a respective longitudinal wall **121**.

[0091] Of course, like prior art pools, the artificial pool **100** in accordance with the present embodiment is intended to have a coating, for example a liner or tiles, and can comprise equipment such as a pumping system, a contraflow generation system (to enable swimming against a current), decompression or drainage pits, a water processing system, etc. Such coatings and equipment being known in the prior art, they are not described as such in the present document.

[0092] Such an artificial pool **100** can be assembled using a production method comprising the following steps: [0093] providing a plurality of panels **110** intended to be arranged along the perimeter of the artificial pool **100** and together to form the lateral walls **101**, **102** of the artificial pool **100**, [0094] providing a plurality of lateral uprights **120** intended to be arranged along the perimeter of the artificial pool **100**, [0095] providing for each lateral wall **101**, **102** of the artificial pool **100** respective first and second crossmembers **131**, **133** intended to be arranged along said lateral wall **101**, **102**, [0096] providing first angle irons **140**, [0097] providing support strut elements **121**, [0098] arranging the panels **110** along the perimeter of the artificial pool **100** and removably fixing the panels **110** together in such a manner that two successive panels **110** are removably fixed to one another and the panels **110** of said plurality of panels together form the lateral walls **101**, **102** of the artificial pool **100**, [0099] arranging the uprights along the exterior perimeter of the artificial pool and removably fixing each lateral upright **120** to at least one panel **110**, said lateral upright **120** preferably being fixed to the at least one panel at the level of a junction with another panel **110**, [0100] arranging each first and each second crossmember **131**, **133** along said corresponding lateral wall **101**, **102** and fixing said first and second crossmembers **131**, **133** to each panel **110** constituting said lateral wall **101**, **102** either via the lateral upright **120** fixed to said panel **110** or via a fixing part **123** removably fixed to said removable panel **110**, the first and second crossmembers **131**, **133** being removably fixed together at the level of the edges by the first angle irons **140** in such a manner as to form first and second frames **132**, **134**.

[0101] Once the artificial pool **100** has been assembled in this way it can be provided with a coating in order to seal it and have the required equipment installed.

[0102] FIG. **2** depicts an artificial pool **100** in accordance with a second embodiment similar to the first embodiment and in which a platform **105** is provided that is housed at least in part in a volume delimited by the at least four lateral walls **101**, **102**, said platform **105**, or mobile bottom, being mobile between a so-called lower first position and a so-called upper second position in such a manner as to modify an apparent depth of said artificial pool **100**.

[0103] Note that in the present document the terms “platform” and “mobile bottom” are used interchangeably, so that the term “mobile bottom” can be substituted for “platform” and vice versa.

[0104] In accordance with one possibility of the invention, the second position can be an out of the water position in which the platform **105** closes the artificial pool **101** and participates in the formation of a patio.

[0105] An artificial pool **100** in accordance with the second embodiment differs from the artificial pool **100** in accordance with the first embodiment in that it comprises a mobile platform **105** and elements associated with that platform, in particular actuating elements.

[0106] Furthermore, in order to depict the modular design possibilities offered by the invention, the various elements of the artificial pool **100** are also depicted in FIGS. **3**A to **6**C, which concern an artificial pool **100** in accordance with a variant of the second embodiment that differs from the artificial pool **100** in accordance with the second embodiment in that it has a length of 6 m instead of 8 m. As for the remainder, apart from the differences linked to this difference in length, notably the fact that the artificial pool **100** comprises only four support strut elements **121**, the artificial pool as depicted in FIGS. **3**A to **6**C has an identical configuration to that of the artificial pool **100** in accordance with the second embodiment. Because of this, in the remainder of this document elements common to the artificial pool **100** in accordance with the second embodiment and its variant are described with reference to FIGS. **2** to **6**C without specifying the embodiment and the variant to which this relates.

[0107] The platform **105** takes the form of a second assembly of beams **105**A and, as depicted in FIGS. **5**D, a covering **105**B such as tiles, PVC (polyvinyl chloride) or wood slats, supported by the second set of beams **105**A.

[0108] In order to render the platform **105** mobile, the artificial pool **100** in accordance with the second embodiment and its variant comprises: [0109] a system **140** for actuating the platform arranged outside the volume delimited by the at least four lateral walls **101**, **102**, [0110] at least four cables **151**, in the present embodiment and its variant six cables **151** shown only in FIG. **2**, connecting the platform **105** to the actuating system **140**, movement of said cables **151** being driven by the actuating system **140**, and [0111] at least four first pulleys **156**, in the present embodiment and its variant six pulleys **156**, arranged along the lateral walls **101**, **102** in such a manner as to guide a respective cable **151** toward traction points **1053** of the platform **105** to apply vertical traction to it.

[0112] As FIG. **4** shows, the actuating system **140** comprises a cylinder **141** the two ends of travel of which respectively correspond to the first position and the second position. To be more specific, the actuating system further comprises: [0113] a casing **143** shaped to be fixed to one of the lateral walls **102** of the artificial pool **100**, the cylinder **141** being fastened to the casing **143**, [0114] a shuttle and pulley system **142** comprising: [0115] a shuttle movement of which is driven by the piston of the cylinder **141** and comprising four secondary pulleys, [0116] four attachments and four other secondary pulleys fastened to the casing opposite the cylinder in such a manner that each cable **151** attached to a respective attachment passes over a respective pulley of the shuttle to then pass over a respective other secondary pulley.

With such a configuration the movement of the cylinder **141** enables duplicate driving of movement of the four cables **151**.

[0117] In order to guide the cables at each end of the platform, in addition to the four first pulleys the artificial pool **100** further comprises the upper frame **153** fixed to the lateral uprights **120**. The upper frame **153** includes two arms to enable fixing of the casing **143** and two longitudinal housings **153**A, **153**B each associated with one of the longitudinal walls **102** to guide the cables **151** in order to feed them to the traction points **1051** of the platform **105**. To this end, as shown in FIGS. **5**B to **5**D, the upper frame **153** comprises: [0118] guide pulleys **154** guiding the cables along the longitudinal housings **153**A, **153**B, and [0119] first pulleys **156** arranged regularly along the upper frame **153**, here along the longitudinal edges near each corner and at a median location on each longitudinal edge, and with their rotation axis perpendicular to the corresponding longitudinal wall in order to guide the corresponding cable **151** from the horizontal direction in the corresponding longitudinal housing **153**A, **153**B to a vertical direction in order to apply traction to the platform **105**.

[0120] In accordance with one possibility of the invention the upper frame **153** can have two longitudinal sides and two lateral sides; each longitudinal side has a section of U general shape accommodating, between the arms of the U, rolling elements such as pulleys. In accordance with this possibility the upper frame **153** is configured to allow: [0121] removable fixing to the panels **110** via the bottom of the U, the rolling elements then being able to provide a support for a sliding cover, not shown, and [0122] removable fixing to the panels **110** via these arms, the artificial pool in accordance with the present embodiment comprising the platform **105** and the rolling elements forming the first pulleys arranged along the lateral walls **101**, **102** in such a manner as to guide the respective cable **151**.

[0123] Such an upper frame **153** offers the possibility of using the same upper frame **153** either for winding and unwinding a sliding cover of the pool or, in accordance with the present embodiment, providing a mobile bottom. It is therefore entirely possible to adapt the configuration of the artificial pool **100** without having to provide a plurality of different upper frames **153**. Note also that in the present embodiment the longitudinal housings **153**A, **153**B correspond to the housing delimited by the arms of the U of the longitudinal sides of the upper frame **153**.

[0124] In the same manner and as depicted in FIGS. **6**A to **6**C, in order to allow vertical traction on the platform during its movement between the first and second positions, the platform **105** comprises: [0125] at least four traction points **1052**, also termed anchor points, to which the cables **151** are intended to be fixed, six such points in the present embodiment and its variant, four traction points **1052** being associated with the four corners of the platform **105** and two other traction points **1052** each being arranged on a median portion of the respective longitudinal edge of the platform **105**, [0126] third pulleys each associated with a traction point **1052** to guide the corresponding cable **151** toward said traction point **1052**, [0127] guide and centering pads **1053** shaped to guide

and center the platform during its movement between the first and second positions.
[0128] Thanks to such a conformation and the guiding of the cables **151** made possible by the various pulleys **1052**, **156**, **153**, the movement of the piston of the cylinder **141**, and therefore of the shuttle, enables pulling on or releasing the traction on the cables **151** and therefore raising or lowering the platform **105** in the artificial pool **100**, thereby enabling the apparent depth thereof to be changed.
[0129] Of course, it will be noted that in configurations already known from the prior art the artificial pool **100** and its platform **105** may comprise equipment conformed to adapt to the movement of the platform **105**, such as fixed benches or steps.

Claims

1. An artificial pool having at least four lateral walls, two successive lateral walls being connected to one another by an edge, the artificial pool comprising: a plurality of panels arranged along the perimeter of the artificial pool, two successive panels being removably fixed to one another in such a manner that the panels of said plurality of panels together form the lateral walls of the artificial pool, a plurality of lateral uprights arranged along the perimeter of the artificial pool (**100**), each lateral upright being removably fixed to at least one panel for each of the lateral walls of the artificial pool at least one respective first crossmember arranged along said lateral wall, said first crossmember being fixed to each panel forming part of said lateral wall either via the lateral upright fixed to said panel or via a fixing part fixed to said panel, wherein the artificial pool further comprises first angle irons, the first crossmembers being removably fixed together at the level of the edges by said first angle irons in such a manner as to form a first frame.
2. The artificial pool as claimed in claim 1, wherein each lateral uprights are fixed to the corresponding at least one panel at the level of a junction with another panel.
3. The artificial pool as claimed in claim 1 wherein the artificial pool further comprises for each lateral wall at least one respective second crossmember arranged along said lateral wall parallel to the first crossmember corresponding to said lateral wall, said second crossmember being fixed to each panel forming part of said lateral wall either via the lateral upright fixed to said panel or via a fixing part (removably fixed to said panel, wherein pool the second crossmembers are connected to one another at the level of the edges (either by the first angle irons or by second angle irons of the artificial pool in such a manner as to form a second frame parallel to the first frame.
4. The artificial pool as claimed in claim 1, wherein each lateral upright is intended to be fixed to the ground.
5. The artificial pool as claimed in claim 1, further comprising at least one support strut element removably fixed to at least one panel.
6. The artificial pool as claimed in claim 5, wherein the artificial pool comprises four lateral walls, two longitudinal walls and two walls transverse to said longitudinal walls, and wherein the artificial pool comprises for each longitudinal wall at least two support strut elements.
7. The artificial pool as claimed in claim 5, comprising for at least one of the lateral walls at least two support strut elements arranged along said lateral wall, and wherein the artificial pool further comprises for said lateral wall at least one secondary crossmember, said secondary crossmember being fixed to said at least two support strut elements.
8. The artificial pool as claimed in claim 1 further comprising a platform accommodated at least in part in a volume delimited by the at least four lateral walls, said platform being mobile between a first and a second position in such a manner as to modify an apparent depth of said artificial pool.
9. The artificial pool as claimed in claim 8, further comprising: a system for actuating the platform arranged outside the volume delimited by the at least four lateral walls, at least four cables connecting the platform to the actuating system, movement of said cables being driven by the actuating system, and at least four pulleys arranged along the lateral walls in such a manner as to

guide a respective cable to apply vertical traction to the platform (**105**), wherein the actuating system.

10. The artificial pool as claimed in claim 9, the actuating system comprises a cylinder the two ends of travel of which respectively correspond to the first position and the second position.

11. The artificial pool as claimed in claim 1 comprising an upper frame fixed to the panels on an upper part of the latter, the upper frame having two longitudinal sides and two lateral sides, each longitudinal and/or lateral side of the upper frame has a section of U general shape accommodating between the arms of the U rolling members such as pulleys, and wherein said frame is configured to allow either one of: removable fixing of the upper frame to the panels by the bottom of the U, the rolling elements then being adapted to provide a support for a sliding cover, and removable fixing of the upper frame to the panels by its arms, wherein, the artificial pool including a platform, rolling elements; for respective cable to apply vertical traction to the platform, being arranged along the lateral walls in such a manner as to guide the respective cable.

12. The artificial pool as claimed in claim 1, wherein the panels are identical and wherein each panel advantageously has: a width between 25 and 100 cm, a height between 50 and 220 cm.

13. The artificial pool as claimed in claim 1, wherein each panel comprises a bent plate at least two longitudinal edges of which are bent, two successive panels of a lateral wall being assembled to one another by means of their longitudinal edges.

14. A method of assembling an artificial pool, the artificial pool having at least four lateral walls, two successive lateral walls being connected to one another by an edge, said method comprising the following steps: providing a plurality of panels intended to be arranged along the perimeter of the artificial pool and together to form the lateral walls of the artificial pool, providing a plurality of lateral uprights intended to be arranged along the perimeter of the artificial pool, providing for each lateral wall of the artificial pool at least one respective first crossmember intended to be arranged along said lateral wall, providing first angle irons, arranging the panels along the perimeter of the artificial pool and removably fixing the panels to one another in such a manner that two successive panels are removably fixed to one another and the panels of said plurality of panels together form the lateral walls of the artificial pool, arranging the lateral uprights along the exterior perimeter of the artificial pool and removably fixing each lateral upright to at least one panel, arranging each first crossmember along said corresponding lateral wall and fixing said first crossmember to each panel constituting said lateral wall either via the lateral upright fixed to said panel or via a fixing part removably fixed to said panel, the first crossmembers being removably fixed to one another at the level of the edges by the first angle irons in such a manner as to form a first frame.